

Emerging Directions in Deep Learning

Examples of ML techniques on particular manifolds (II)

Mihai-Sorin Stupariu, 2025-2026

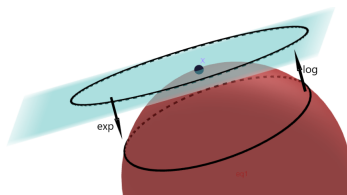
Recap & key points

The projective plane and the projective space

The Stiefel and the Grassmann manifolds

Recap: several key concepts

- ▶ Manifold
- ▶ Tangent space
- ▶ Riemannian metric on a manifold; Riemannian manifold
- ▶ Exponential map, logarithm map (locally defined, depend on the point!), these maps establish a correspondence between the neighborhood of any point on a Riemannian manifold and its tangent space.



- ▶ Geodesic

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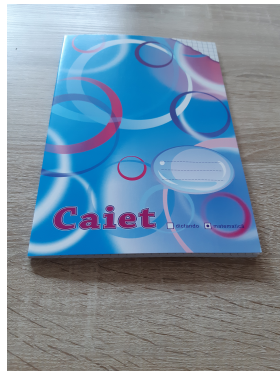
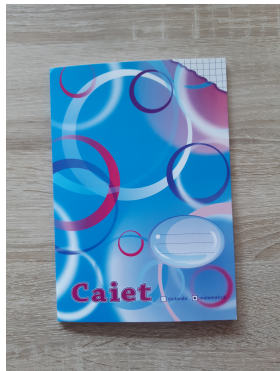
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- ▶ Last course: examples of manifolds of matrices. For instance:

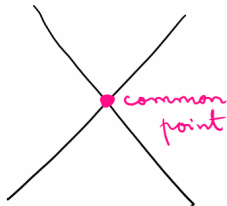
$$\mathrm{SO}(n) \subset \mathrm{O}(n) \subset \mathrm{GL}(n, \mathbb{R}) \subset \mathbb{R}^{n^2}.$$

In particular, one gets an embedding in an Euclidean space, which allows to handle the elements of the manifold as numerical data. Another example was represented by Lie groups (internal algebraic operation, allowed the definition of convolution-like operations).

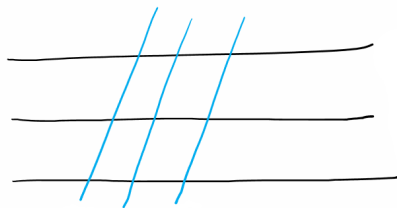
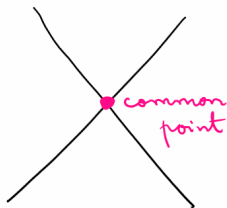
Motivation: tasks in Computer Vision



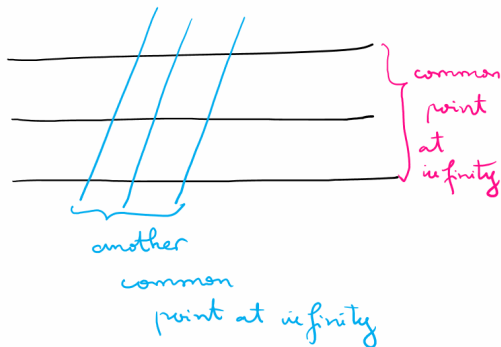
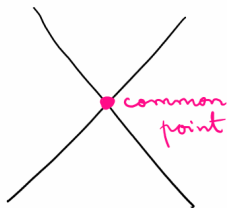
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- ▶ It is hard to “intuitively represent” the real projective plane. The reason: it does not admit an embedding in the Euclidean space $\mathbb{R}^3$. Other examples of surfaces that do not admit such an embedding in $\mathbb{R}^3$ (but only in higher dimensional Euclidean spaces): the Möbius strip (infinite); the Klein bottle.

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- ▶ Can you describe the real projective line $\mathbb{P}^1\mathbb{R}$? The complex projective line $\mathbb{P}^1\mathbb{C}$?

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- ▶ The equivalence class of an element $X = (X_1, X_2, X_0)$ is denoted by $[X_1 : X_2 : X_0]$. Thus, we have $[1 : 2 : -5] = [2 : 4 : -10]$. On the other hand, $[1 : 2 : -5] \neq [1 : 2 : 5]$ and $[1 : 2 : -5] \neq [-5 : 2 : 1]$.

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- ▶ **Terminology.** For an element $\xi \in \mathcal{C}$, if $\xi = [X_1 : X_2 : X_0]$, then (X_1, X_2, X_0) are called **the homogeneous coordinates of ξ** . The homogeneous coordinates are defined and are unique up to the multiplication with a non-zero number.

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- **Conclusion:** There exists a natural 1:1 correspondence between the real projective plane $\mathbb{P}^2\mathbb{R}$ (the geometric construction) and the set $\mathcal{C}$ of equivalence classes (the algebraic construction). This means that in $\mathbb{P}^2\mathbb{R}$ one can use the homogeneous coordinates.

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 - ▶ The projective space can be regarded as a **quotient** space:
 $\mathbb{P}^n\mathbb{R} = \mathbb{R}^{n+1} \setminus \{0\} / \mathbb{R}^* \simeq S^n / \mathbb{Z}_2$.

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- ▶ Consider the homographies given by the matrices

$$M_1 = \begin{pmatrix} 2 & 3 & 4 \\ 6 & 5 & -1 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 2 & -1 & 4 \\ 6 & -4 & 8 \\ 3 & 1 & 2 \end{pmatrix}.$$

The homography corresponding to M_1 leaves the line at infinity ($X_0 = 0$) unchanged. The homography corresponding to M_2 changes the position of the line at infinity. In particular, parallel lines are transformed into concurrent lines.

Examples (I)

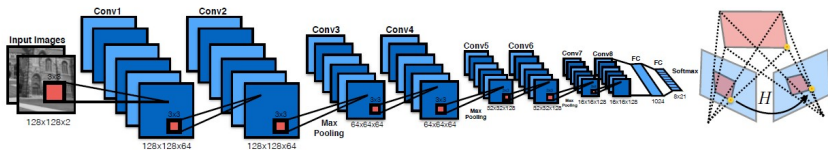


Fig. 1: **Deep Image Homography Estimation.** HomographyNet is a Deep Convolutional Neural Network which directly produces the Homography relating two images. Our method does not require separate corner detection and homography estimation steps and all parameters are trained in an end-to-end fashion using a large dataset of labeled images.

DeTone et al., 2016 - architecture of a deep CNN aiming to generate the homography that relates two images.

Examples (II)

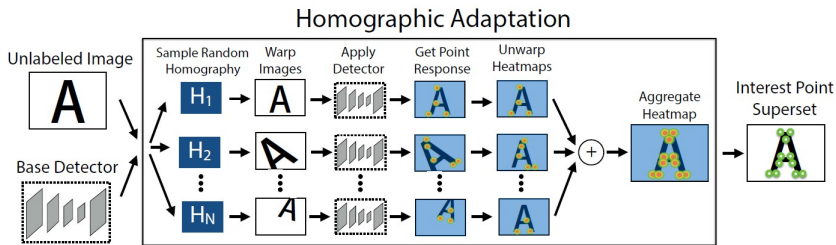


Figure 5. **Homographic Adaptation.** Homographic Adaptation is a form of self-supervision for boosting the geometric consistency of an interest point detector trained with convolutional neural networks. The entire procedure is mathematically defined in Equation 10.

DeTone et al., 2018 - homographies used for boosting the geometric consistency of an interest point detector.

Examples (III)



Fig. 1: Starting from a single spherical panorama, we exploit the combination of geometry (accurate lines) and deep learning (edge map) to recover the main structure of the room, achieving 3D complex layouts.

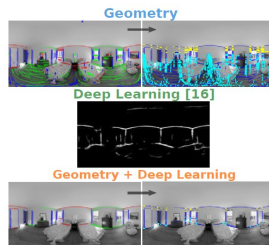


Fig. 3: **Top:** Oriented lines and corners extracted just with geometric reasoning. **Center:** Edge Map obtained through [16]. **Bottom:** Resulting structural lines and corners after combining geometry and deep learning. A large reduction of them is shown, while those more significant for the main structure remain. Corners become good candidates for the hypotheses generation.

Fernandez-Labrador et al., 2018 - the geometry of projective planes is taken into account for 3D layout recovery of indoor scenes.

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- ▶ An example: [Lee et al., 2003](#) - for video-face recognition. Steps: (i) each person is represented by a manifold embedded in the ambient Euclidean space; (ii) one gets a collection of subsets (called pose manifolds); (iii) each pose manifold is approximated by a linear space (actually a plane) - need to work in the space of all such planes in $\mathbb{R}^3$.
This is actually a manifold - the Grassmann manifold!

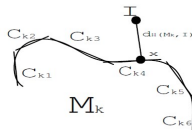


Figure 1: Appearance manifold. A complex and nonlinear manifold can be approximated as the union of several simpler pose manifolds; here, each pose manifold is represented by a PCA plane.

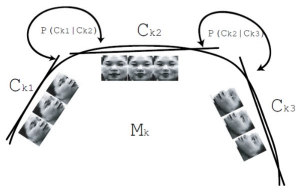


Figure 3: Dynamics among pose manifolds. The dynamics among the pose manifolds are learned from training videos which describes the probability of moving from one manifold to another at any time instance.

The Stiefel manifold and the Grassmann manifold - definition

- Consider the set of k -orthonormal frames in $\mathbb{R}^n$ ($1 \leq k \leq n$).

$$V_k(\mathbb{R}^n) = \text{St}(k, n) = \{(w_1, \dots, w_k) \mid w_i \in \mathbb{R}^n, \langle w_i, w_j \rangle = \delta_{ij}, \forall i, j\}.$$

The set $V_k(\mathbb{R}^n)$ can be endowed in a natural manner with a manifold structure of dimension $nk - \frac{k(k+1)}{2}$. It is called the **Stiefel manifold**. For instance, if $k = 1$, one has $V_1(\mathbb{R}^n) \simeq S^{n-1}$.

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- Both manifolds have Geomstats implementations: the [Stiefel manifold](#) and the [Grassmann manifold](#).

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- ▶ The vectors $w_1 = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T, w_2 = (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T$ yield the 2-orthonormal frame $\mathbf{w} = (w_1, w_2) \in V_2(\mathbb{R}^3)$. One can associate to $\mathbf{w}$ the matrix

$$M_{\mathbf{w}} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 \end{pmatrix}.$$

The Stiefel manifold and the Grassmann manifold - matrix representations

- ▶ **Example.** Take $n = 3, k = 2$.
- ▶ The vectors $u_1 = (1, 0, 0)^T, u_2 = (0, 1, 0)^T$ yield the 2-orthonormal frame $\mathbf{u} = (u_1, u_2) \in V_2(\mathbb{R}^3)$. One can associate to $\mathbf{u}$ the matrix

$$M_{\mathbf{u}} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

- ▶ The vectors $w_1 = (\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T, w_2 = (-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0)^T$ yield the 2-orthonormal frame $\mathbf{w} = (w_1, w_2) \in V_2(\mathbb{R}^3)$. One can associate to $\mathbf{w}$ the matrix

$$M_{\mathbf{w}} = \begin{pmatrix} \frac{\sqrt{2}}{2} & -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{2} \\ 0 & 0 \end{pmatrix}.$$

- ▶ The vectors $t_1 = (0, 0, 1)^T, t_2 = (0, 1, 0)^T$ yield the 2-orthonormal frame $\mathbf{t} = (t_1, t_2) \in V_2(\mathbb{R}^3)$.

The Stiefel manifold and the Grassmann manifold - matrix representations

► **Example.** (continued)

The Stiefel manifold and the Grassmann manifold - matrix representations

- ▶ **Example.** (continued)
- ▶ Denote by (x, y, z) the coordinates in $\mathbb{R}^3$. Consider the plane $\pi : z = 0$, thus $\pi \in \text{Gr}(2, 3)$. One can **choose** a frame of orthogonal vectors with norm 1 that span π (such a frame is not unique!).

The Stiefel manifold and the Grassmann manifold - matrix representations

- ▶ **Example.** (continued)
- ▶ Denote by (x, y, z) the coordinates in $\mathbb{R}^3$. Consider the plane $\pi : z = 0$, thus $\pi \in \text{Gr}(2, 3)$. One can **choose** a frame of orthogonal vectors with norm 1 that span π (such a frame is not unique!).
- ▶ With the notations used on the previous slide, one has

$$\text{Span}(u_1, u_2) = \pi, \quad \text{Span}(w_1, w_2) = \pi, \quad \text{but} \quad \text{Span}(t_1, t_2) \neq \pi.$$

The Stiefel manifold and the Grassmann manifold - matrix representations

- ▶ **Example.** (continued)
- ▶ Denote by (x, y, z) the coordinates in $\mathbb{R}^3$. Consider the plane $\pi : z = 0$, thus $\pi \in \text{Gr}(2, 3)$. One can **choose** a frame of orthogonal vectors with norm 1 that span π (such a frame is not unique!).
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- ▶ One can choose the matrix M_u as a representative of π . However, this matrix representation is not unique! The matrix M_w is also a representative of π . One has $M_u = M_w \cdot Q$, with $Q \in O(2)$.

The Stiefel manifold and the Grassmann manifold

TABLE 2.1
Representations of subspace manifolds

SPACE	SYMBOL	MATRIX REP.	QUOTIENT REP.
Orthogonal Group	O_n	Q	—
Stiefel Manifold	$V_{n,p}$	Y	O_n/O_{n-p}
Grassmann Manifold	$G_{n,p}$	None	$\left\{ \begin{array}{c} V_{n,p}/O_p \\ \text{or} \\ O_n/(O_p \times O_{n-p}) \end{array} \right\}$

Source: [Edelman et al., 1998](#)

Applications (I)

Activity	Dim. Red. [31] 16^3 volume	Best Dim. Red. [31] 64^3 volume	Subspace Angles 16^3 volume	NN-Pro- Stiefel 16^3 volume	Kernel- Stiefel 16^3 volume
Check Watch	76.67	86.66	93.33	90	100
Cross Arms	100	100	100	96.67	100
Scratch Head	80	93.33	76.67	90	96.67
Sit Down	96.67	93.33	93.33	93.33	93.33
Get Up	93.33	93.33	86.67	80	96.67
Turn Around	96.67	96.67	100	100	100
Walk	100	100	100	100	100
Wave Hand	73.33	80	93.33	90	100
Punch	83.33	96.66	93.33	83.33	100
Kick	90	96.66	100	100	100
Pick Up	86.67	90	96.67	96.67	100
Average	88.78	93.33	93.93	92.72	98.78

Table 1. Comparison of view invariant recognition of activities in the INRIA dataset using a) Best DimRed [31] on $16 \times 16 \times 16$ features, b) Best Dim. Red. [31] on $64 \times 64 \times 64$ features c) Nearest Neighbor using Procrustes distance on the Stiefel manifold ($16 \times 16 \times 16$ features), d) Maximum likelihood using kernel density methods on the Stiefel manifold ($16 \times 16 \times 16$ features)

Turaga et al., 2008 - statistics on Stiefel and Grassmann manifolds with applications in Computer Vision

Applications (II)

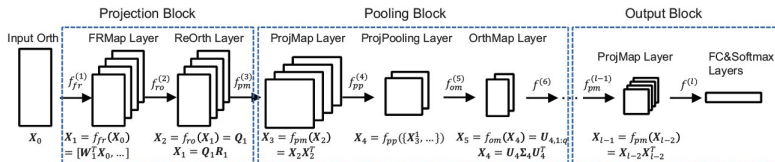


Figure 1: Conceptual illustration of the proposed Grassmann Network (GrNet) architecture. The rectangles in blue represent three basic blocks, i.e., Projection, Pooling and Output blocks, respectively.

Huang et al., 2018 - deep network architecture by generalizing the Euclidean network paradigm to Grassmann manifold

Applications (III)

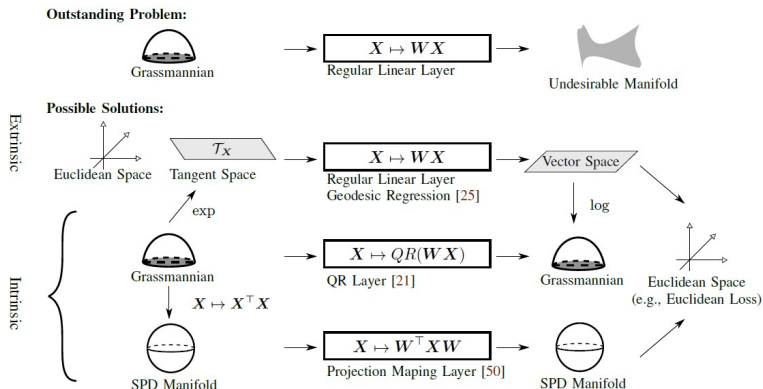


Fig. 10: The problem of existing neural networks and different approaches for constructing mapping network layers on the Grassmann manifold.

Zhang et al., 2018 - approaches for network design on the Grassmann manifold

Other applications

- ▶ [Hamm and Lee, 2008](#) - Grassmann Kernels for subspace-based learning problems

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- ▶ [Harandi et al., 2013](#). The paper deals with sparse dictionary learning over the space of linear subspaces (i.e. the Grassmann manifold). The authors use an embedding of the manifold in a space of matrices.

Other applications

- ▶ [Hamm and Lee, 2008](#) - Grassmann Kernels for subspace-based learning problems
- ▶ [Harandi et al., 2013](#). The paper deals with sparse dictionary learning over the space of linear subspaces (i.e. the Grassmann manifold). The authors use an embedding of the manifold in a space of matrices.
- ▶ [Som et al., 2018](#). The paper constructs probability density functions that capture topological noise of a given input data and that yields a subspace structure, that is a point on a Grassmann manifold.

Main theoretical references

Edelman et al., 1998

Absil et al., 2004

Zimmermann, 2017

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Further theoretical references

B. Conrad, Course Notes - Gluing Topologies

B. Conrad, Course Notes - Quotients by group actions

Henkel, 2005